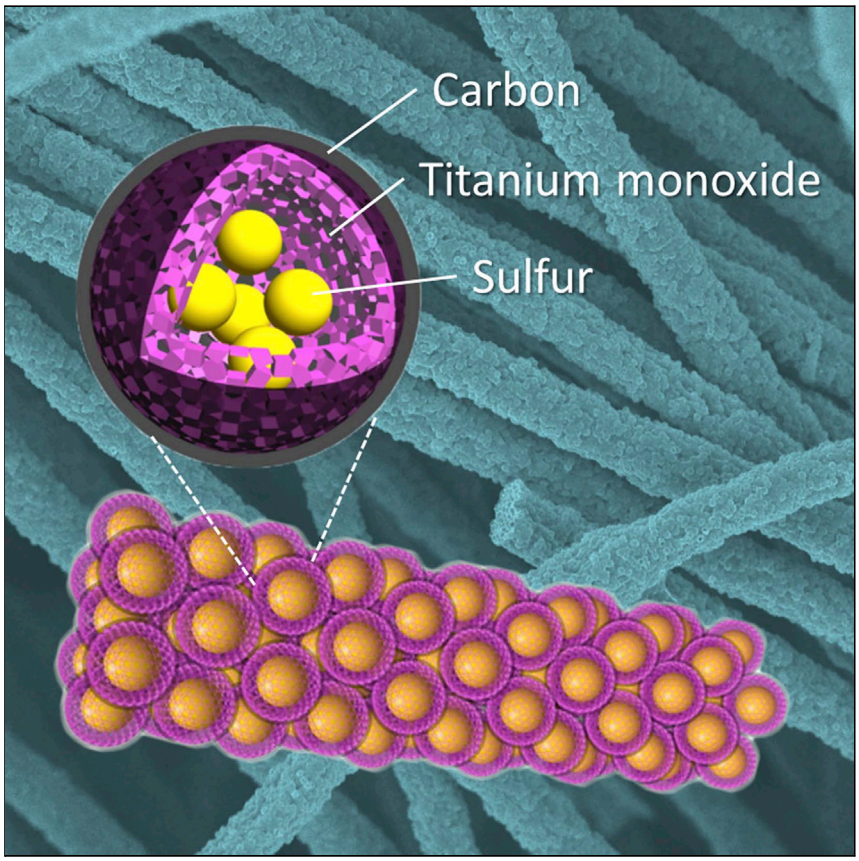
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A Compact Nanoconfined Sulfur Cathode for High-Performance Lithium-Sulfur Batteries



This work proposes a hierarchically structured cathode that simultaneously tackles several problems associated with high-sulfur-loading electrodes for lithium-sulfur batteries. This work overcomes the major limitations associated with other host materials of sulfur, and opens up new prospects for constructing more efficient nanostructures for moderating the diffusion loss of polysulfides and enhancing the reaction kinetics of sulfur. We hope this work will inspire scientists to develop better batteries to satisfy the world demand for energy storage.

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HIGHLIGHTS

TiO hollow nanospheres are closely packed into "clusters" for loading S

The micrograde fibrous cathode can form efficient high-mass-loading film

The Li-S battery enables superior cycle and rate performance

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A Compact Nanoconfined Sulfur Cathode for High-Performance Lithium-Sulfur Batteries

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SUMMARY

A high-sulfur-loading cathode is the most essential component for lithium-sulfur batteries to gain considerable energy density for practical applications. The main challenges associated with high-sulfur-loading electrodes are low specific capacity caused by the insulating nature of sulfur and poor stability arising from dissolution of polysulfides into most organic electrolytes. Here, we propose a hierarchically structured sulfur cathode that simultaneously tackles both problems associated with high-sulfur-loading electrodes. Specifically, titanium monoxide hollow nanospheres are packed space efficiently and closely connected by carbon layers into microsized "clusters" as the sulfur host. As a result of this hierarchical structure design, the nanoscale reaction chambers and the microscale conductive networks cooperatively promise high capacities of sulfur at various current densities. During cycling, the titanium monoxide polar layer in every nanocompartment effectively hinders the dissolution of polysulfides, resulting in superior cycling stability for the high-sulfur-loading electrode.

INTRODUCTION

The lithium-sulfur (Li-S) battery is considered one of the most promising candidates for next-generation electric energy storage due to its high energy density and low material cost, both of which are very important for the emerging large-scale applications such as electric vehicles and grid energy storage.[1–3] The main challenges associated with Li-S batteries are low specific capacity caused by the insulating nature of sulfur, and poor stability arising from dissolution of polysulfides into most organic electrolytes. In the last few years, numerous efforts have been devoted to solving these problems with great progress achieved in many aspects,[4] such as composing sulfur with conductive hosts,[5–8] restricting the dissolution of polysulfides through chemical bonding,[9–14] developing new concept electrolytes,[15–18] and designing novel cell structures.[19–23] Benefiting from these efforts, the capacity of sulfur has increased to >1,000 mAh g^{-1}, and the cycle life of Li-S batteries has been prolonged to 500 cycles or more.[24–26] However, although the electrochemical performance of Li-S batteries has been significantly improved, many reported good properties are achieved using light coated electrodes with areal sulfur loading less than 2.0 mg cm^{-2}, which deliver actual areal capacities far less than values for the state-of-art lithium-ion batteries.[27,28] Unfortunately, when the mass loading of sulfur on electrodes is increased to a certain level, the above-mentioned problems become significant again. It is generally understood that electrodes with low areal mass loading and low sulfur content, even with superior electrochemical properties, are still far from commercialization.[28–32] Therefore, to bring Li-S batteries closer to practical application, the development

Context & Scale

The number of mobile phones in the world is forecasted to exceed 5 billion in the next few years, and the marketing of electric vehicles is quickly rising worldwide. Growing along these emerging applications is the demand for long-lasting and cost-effective batteries to power not just smartphones in our hands but also the electric vehicles on the streets. Due to their high capacity and low cost, lithium-sulfur (Li-S) batteries are seen as promising candidates for next-generation electrical energy storage. However, current systems are hampered by low capacity mainly restricted by the insulating nature of S and poor stability due to the dissolution of polysulfides. Here, we propose a hierarchically structured cathode that simultaneously tackles both problems associated with the high-S-loading electrodes. We hope that this work might inspire scientists to develop high-efficiency S cathodes, so that Li-S batteries can be realized in our daily life in the near future.

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of high-mass-loading cathodes with high sulfur content is now one of the most urgent research tasks for Li-S batteries.

Some groups have paid attention to the above-mentioned issues and have done several studies focusing on increasing the mass loading on electrode films.[21,22,33–37] The sulfur loadings of some electrodes have been remarkably enhanced to >5.0 mg cm^{-2}, or even higher than 10 mg cm^{-2}, without sacrificing too much specific capacity of sulfur.[21,33,38] Nonetheless, it is interesting to note that many of the previously reported high-sulfur-loading electrodes are based on the integral carbon-based structures, such as carbon nanofiber-woven papers,[21,33,39] graphene foam-supported films,[37,40] and porous carbon interlayers.[19,36] The self-supported structures reduce the usage of polymer binders and increase the sulfur content of the electrode. However, these film electrodes still manifest mostly fast capacity decay, and their cycle life is usually less than 200 cycles.[33,36,40] This is because the substrates usually cannot provide effective adsorption of the dramatically increased polysulfides in high-sulfur-loading electrodes. Another observation is that the manufacture of the self-supported integral electrodes is not cost effective for mass production. More importantly, such film electrodes, while promising for microbatteries, are not really compatible with the current manufacturing protocol used in battery plants today. Thus, the development high-mass-loading electrodes that can be achieved through the normal slurry coating method will be more compatible and meaningful for realistic engineering developments.[27]

Compared with micrograde sulfur particles, nanosized sulfur has many advantages for achieving enhanced properties due to the size effect.[7,41] Moreover, the design and synthesis of hollow nanostructures as sulfur hosts can remarkably improve the electrochemical performance of Li-S batteries.[9] Recently, we reported titanium monoxide@carbon (TiO@C) hollow nanospheres as an advanced sulfur host for Li-S batteries.[42] Benefiting from high conductivity and effective capture of polysulfides, the nanoparticle-type composite cathode (denoted as NP-TiO@C/S) delivers a high capacity of >1,100 mAh g^{-1} accompanied by a prolonged cycle life of 500 cycles. However, to evaluate the performance of high-loading electrodes with 4.0 mg cm^{-2} of sulfur, the content of conductive additive (Super-P) has to be increased from 10 wt % to 20 wt % to build more efficient conductive networks in the thick-coated electrode film. Similar problems are also found in many previously reported studies. For example, Cui's group added ≥30 wt % of conductive carbon black (Super-P) in Li_2S electrode films.[43,44] Nazar's group used graphene to replace part of Super-P when the mass loading of sulfur was increased,[45,46] and many nanosized cathodes even with low sulfur loading of only 1.0 mg cm^{-2} need to use 20 wt % of conductive additives for the electrodes.[41,47–51] The overuse of conductive additives will inevitably decrease the sulfur content of the electrode and lower the energy density of the whole Li-S battery. Based on the research and development experiences of commercial Li-ion batteries, micrometer-sized active materials are preferred rather than nanosized ones; one of the main reasons is that bigger grains need less polymer binder and conductive additives to construct efficient electrode films.[1] The requirements of particle sizes for higher reaction kinetics and more practical electrodes are apparently conflicting with each other, and the compromise is not easy.

Here, we report a hierarchically structured sulfur cathode that simultaneously tackles all problems associated with the high-mass-loading electrodes. Inspired by the grape cluster structure, titanium monoxide (TiO) hollow nanospheres as "grape grains" are packed space efficiently and closely connected by a carbon sheath

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into microsized "clusters" as the sulfur host. In the proposed host structure, the carbon framework performs connecting and conducting functions, and TiO plays an important role in limiting the loss of polysulfides to the organic electrolyte. As a result of this unique structure, when 73 wt % of sulfur is loaded within the grape-cluster-like titanium monoxide@carbon hollow fiber (GC-TiO@CHF), the nanoscale reaction chambers and the microscale conductive networks cooperatively promise a high capacity of sulfur at various current densities. During charge/discharge processes, the TiO polar layer in every "grape grain" effectively hinders the dissolution of polysulfides, enabling superior cycling stability over 400 cycles for the high-sulfur-loading electrode with 5.0 mg cm^{-2}. In addition, the micrograde fibrous cathode can form an efficient high-mass-loading film on an aluminum foil current collector without using any special binder or additive, which is entirely compatible with conventional battery manufacturing protocols.

RESULTS

Inspired by the unique structure of the grape cluster (Figure 1A), this work aims to construct a micrograde hierarchically structured sulfur host for advanced Li-S batteries. To synthesize the designed cathode material (Figure 1B), SiO_2 nanospheres were coated with a layer of TiO_2 precursor, and then electrospun with polyacrylonitrile (PAN) into SiO_2@TiO_2@PAN fibers. After the annealing and etching steps, the GC-TiO@CHF host was gained by a further annealing process at 1,000°C in an N_2:H_2 (95:5) atmosphere. Finally, the GC-TiO@CHF/S composite was obtained by loading sulfur within the hollow host. By reducing the diameter of the reaction chamber and optimizing their packing arrangement, the electrochemical reaction kinetics of sulfur can be significantly enhanced due to the size effect (Figure S1). Thus, although the previously reported NP-TiO@C/S cathode shows acceptable C-rate properties in a high-sulfur-loading electrode,[42] the GC-TiO@CHF/S cathode in this work will significantly enhance the comprehensive electrochemical performance of Li-S batteries. Uniform SiO_2 nanospheres, with an average diameter of ~165 nm, were prepared as the hard template for creating void space to accommodate sulfur (Figures 1C, 1G, and S2). A layer of TiO_2 precursor was uniformly and firmly coated on SiO_2 nanospheres through a facile method reported previously (Figures 1D, 1H, and S3).[52] The average diameter of the SiO_2@TiO_2 nanospheres increased to about 230 nm, suggesting that the thickness of TiO_2 sheath was around 32 nm, which can be identified by transmission electron microscopy (TEM) (Figure 1H). Through the electrospinning method, the SiO_2@TiO_2 nanospheres were closely assembled together with PAN, forming micrometer-sized fibers with diameter in the range of 1.1–1.6 μm and length of hundreds of micrometers (Figures 1E, 1I, and S4). The spherical feature of SiO_2@TiO_2 can be recognized on the surface of the SiO_2@TiO_2@PAN fiber. After annealing at 700°C for 3 hr in an N_2 atmosphere, rutile TiO_2 was formed (Figure S5), which can resist the corrosion of dilute HF acid. SiO_2 nanospheres were completely etched by immersing in 4 wt % HF for 8 hr, generating nanosized void spaces inside every "grape grain" of the whole "cluster" (Figures 1F, 1J, and S6A). GC-TiO_2@CHF maintained the morphology of its precursor and the hollow space can be clearly observed at the cross-section of the fiber (Figures S6B–S6D).

Based on our previous study, the TiO_2 phase can be reduced to TiO phase by high-temperature reduction.[42] After annealing at 1,000°C for 4 hr in an N_2/H_2 (95:5) atmosphere, GC-TiO@CHF maintains its grape-cluster-like hierarchical morphology (Figures 2A and 2B). Nearly 100 nanosized spherical void spaces can be clearly observed at the fracture surface of the fiber, and they are packed in a space-efficient

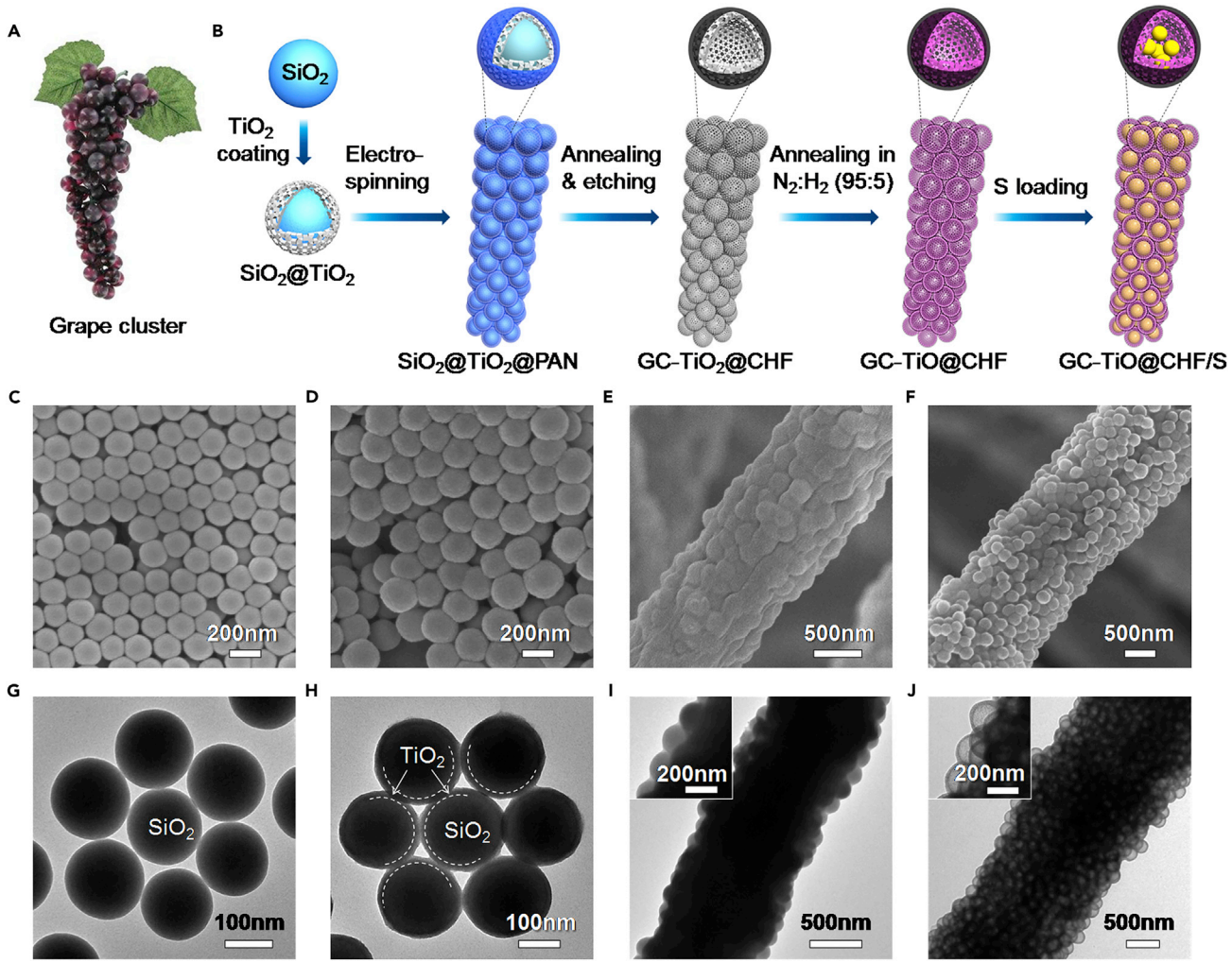


Figure 1. Schematic of the Grape-Cluster-Inspired Design and Characterization of Intermediate Materials

(A and B) Photograph of a grape cluster (A), and schematic illustration of the synthesis process of the GC-TiO@CHF/S cathode (B).

(C–J) SEM (C, D, E, and F) and TEM (G, H, I, and J) images of SiO_2 (C and G), SiO_2@TiO_2 (D and H), SiO_2@TiO_2@PAN (E and I), and GC-TiO_2@CHF (F and J). See also Figures S2–S6.

arrangement (Figure 2C), which can effectively lower the inter-particle resistance. The X-ray diffraction (XRD) pattern reveals the successful formation of the TiO crystal phase (Figure 2D). TEM observations show the hierarchically hollow structures of GC-TiO@CHF (Figures 2E and 2F). A layer of TiO with thickness of several nanometers appears inside the carbon sheath (Figure 2G). The distinct selected area electron diffraction (SAED) pattern further confirms the crystal phase of TiO (Figure 2H). The weight ratio of TiO in GC-TiO@CHF was determined to be about 61 wt % by thermogravimetric analysis (TGA) (Figure S7). Nitrogen adsorption/desorption isotherms reveal that GC-TiO@CHF has a macroporous structure with a specific surface area of 140.5 $m^2\ g^{-1}$ (Figure S8). To demonstrate the essential function of the TiO layer for enhancing cycling stability, a hierarchically hollow carbon host with the similar grape cluster structure (denoted as GC-CHF) was prepared as a control group (Figures 2I–2L). The nanoparticle TiO@C (NP-TiO@C) sample from our previous work was also applied as another control group to show the advantages of the micrograde hierarchical cathode material (Figure S9).[42]

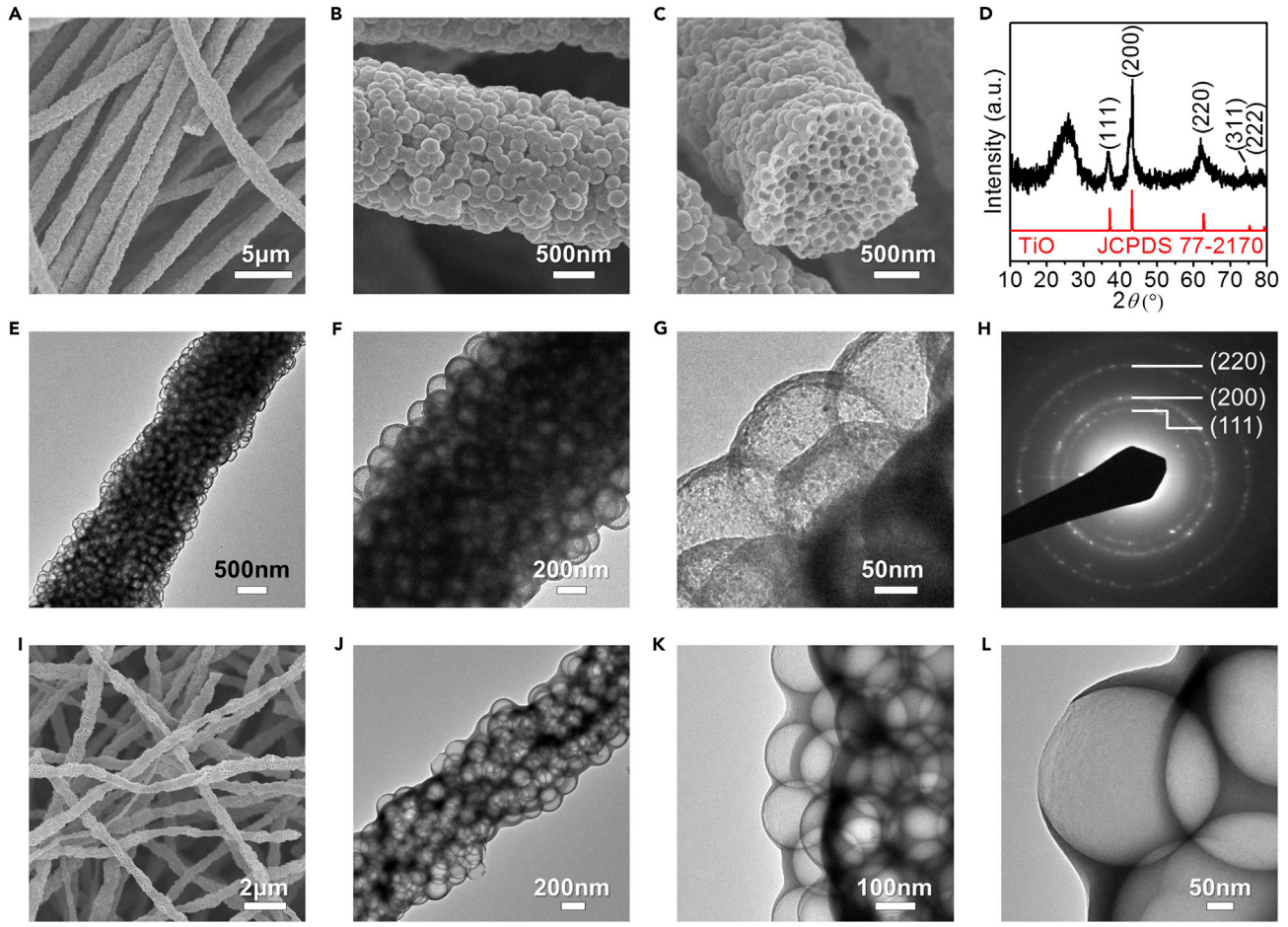


Figure 2. Characterization of Sulfur Host Materials

(A–H) SEM images (A, B, and C), XRD pattern (D), TEM images (E, F, and G), and corresponding SAED (H) pattern of GC-TiO@CHF. (I–L) SEM (I) and TEM images (J, K, and L) of GC-CHF.

A modified vapor phase infusion method was applied to load sulfur into the GC-TiO@CHF host. Scanning electron microscopy (SEM) images show that the morphology of the host was not affected by the sulfur infusing process, and the as-prepared GC-TiO@CHF/S composite maintains the original grape cluster structure (Figures 3A and 3B). The smooth surface of the exposed nanospheres indicates no extra sulfur adhered outside the host (Figure 3C). TEM observations show that the contrast of inner void spaces of GC-TiO@CHF becomes much darker, which reveals the successful accommodation of sulfur within the hollow host (Figures 3D and 3E). Energy dispersive X-ray spectroscopy (EDX) elemental mapping of GC-TiO@CHF/S not only shows the homogeneous encapsulation of sulfur within the host but also further confirms the uniform distribution of TiO in the carbon-connected host (Figures 3F and 3G). The EDX linear scan across one GC-TiO@CHF/S fiber demonstrates that there is a high content of S inside the host (Figures 3F and 3H), further proving that sulfur has infused into the innermost part of the host as expected. The TGA result reveals that a high content of sulfur (73 wt %) was loaded (Figure 3I). The compaction density of GC-TiO@CHF/S was measured to be 1.22 g cm^{-3}. The control group of GC-CHF/S with a sulfur content of 75 wt % was prepared by the same method (Figure S10).

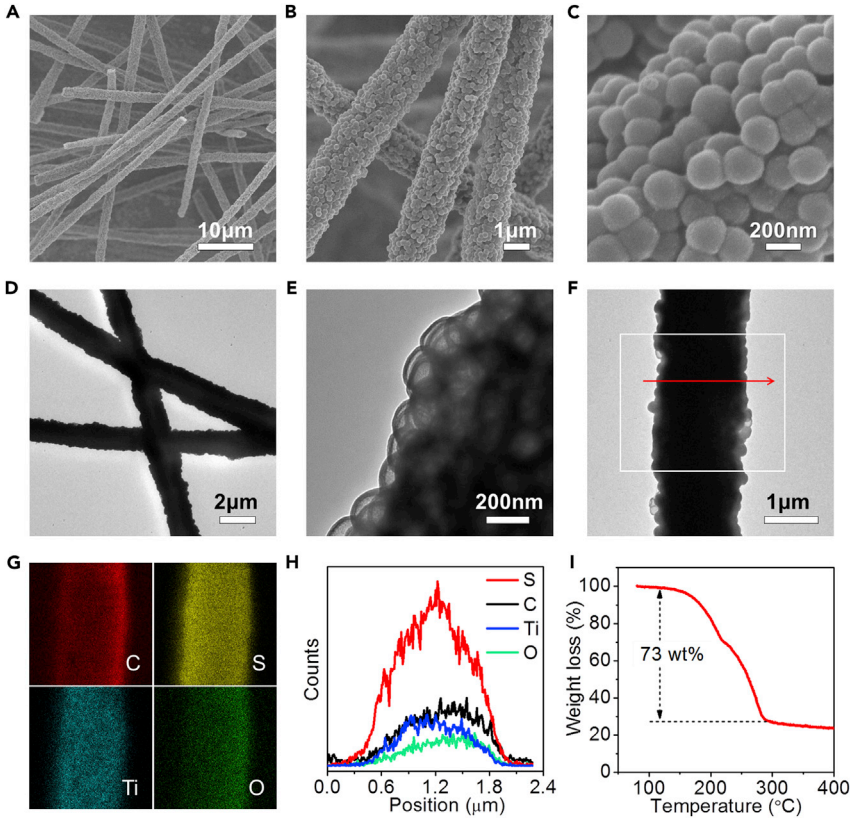


Figure 3. Characterization of GC-TiO@CHF/S

(A–H) SEM (A, B, and C) and TEM (D, E, and F) images of the GC-TiO@CHF/S composite, EDX elemental mappings (G) and linear elemental distributions (H) of C, S, Ti, and O, corresponding to the white square and red arrow in (F), respectively.

(I) TGA curves of the GC-TiO@CHF/S composite under an N_2 atmosphere with a heating rate of 10°C min^{-1}.

To make this cathode material more compatible with commercial battery manufacturing protocols, the electrode films with areal sulfur loading of 5.0 mg cm^{-2} were achieved by traditional slurry coating on an Al foil current collector. The ratio of GC-TiO@CHF/S, conductive carbon (Super-P), and binder (polyvinylidene fluoride [PVDF]) is 8:1:1, which is the most frequently used mass proportion in battery laboratories. Benefiting from the micrograde grain size and one-dimensional morphology, the slurry coating-derived GC-TiO@CHF/S film adheres well to the current collector without any visible fracture issues (Figure S11A). Moreover, with Super-P and the GC-TiO@CHF fiber host, an interconnected conductive network is successfully constructed in the electrode film (Figures S11B and S11C).

The grape cluster design affords impressive improvement on high rate properties for Li-S batteries beyond the nanoparticle-type cathode. As shown in Figure 4A, GC-TiO@CHF/S delivers a high reversible capacity of >1,100 mAh g^{-1} (5.5 mAh cm^{-2}, the value in parentheses corresponds to the areal capacity of the electrode) at 0.05 C, and in the following cycles, it further delivers discharge capacities of 1,000 (5), 850 (4.25), 800 (4), 720 (3.6), and 620 mAh g^{-1} (3.1 mAh cm^{-2}) at 0.1, 0.2, 0.5, 1, and 2 C, respectively (Figures 4A and 4B). On the contrary, even with 20 wt % of Super-P in the electrode film, the NP-TiO@C/S electrode delivers obviously lower capacities of about 950 (4.75), 810 (4.05), 690 (3.45), and

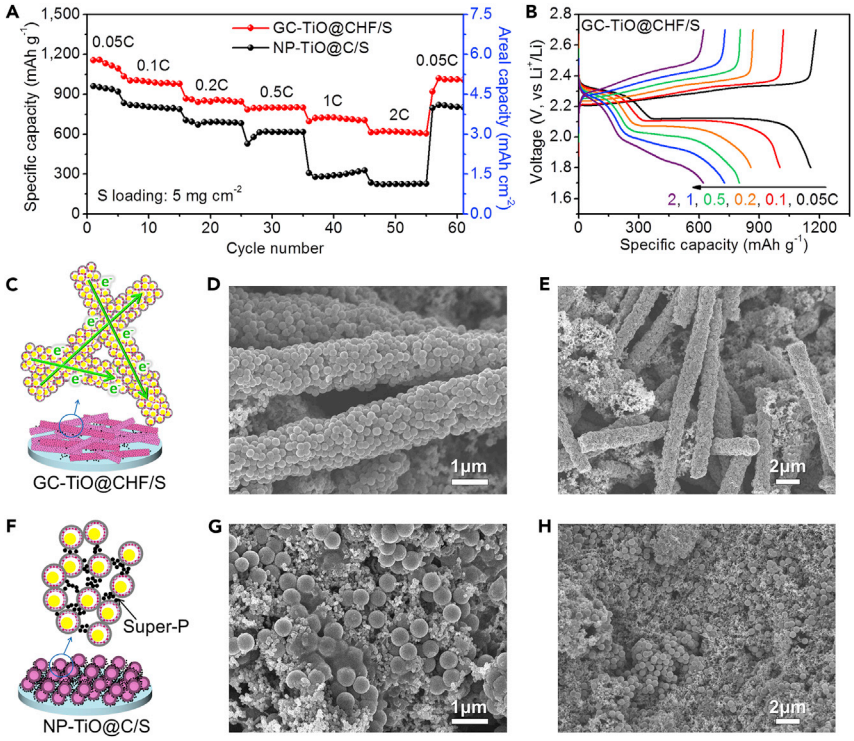


Figure 4. Rate Properties of the GC-TiO@CHF/S Cathode

(A) Comparison of C-rate properties between GC-TiO@CHF/S and NP-TiO@C/S.

(B) Voltage profiles of GC-TiO@CHF/S at various current densities from 0.05 C to 2 C.

(C–E) Schematic (C) and SEM images (D and E) of the GC-TiO@CHF/S electrode.

(F–H) Schematic (F) and SEM images (G and H) of the NP-TiO@C/S electrode.

620 mAh g^{-1} (3.1 mAh cm^{-2}) at 0.05, 0.1, 0.2 and 0.5 C, respectively. When the current density was increased to 1 C, the discharge capacity of NP-TiO@C/S dropped abruptly to around 300 mAh g^{-1} (1.5 mAh cm^{-2}) (Figure 4A). The results imply that, although NP-TiO@C/S can deliver good C-rate capacities based on common low-sulfur-loading electrodes (<2 mg cm^{-2}), its conductive network by carbon additive is not sufficient when the loading mass is increased to a higher level. The Nyquist plots show that GC-TiO@CHF/S has a smaller semicircle in the high-frequency region, indicating smaller charge transfer resistance compared with NP-TiO@C/S (Figure S12). To gain better understanding of the structure-property relationship of the sulfur cathode, the morphologies of the GC-TiO@CHF/S and NP-TiO@C/S electrode films were compared (Figures 4C–4H). When sulfur was incorporated within the host structures, the inner size of the hollow units determined the particle size of active sulfur species to some extent (Figure S1). In the grape-cluster-like structure, the size of the "grape grain" reaction unit was reduced to ~165 nm (Figure 4D), which is around only one-third of the NP-TiO@C/S particle size (Figure 4G). This leads to significantly accelerated redox kinetics of sulfur species due to the size effect. Moreover, compared the traditional way of filling carbon additives in gaps between nanoparticles (Figures 4F–4H), the space-efficient packing arrangement of GC-TiO@CHF/S promises much better electron conduction efficiency (Figures 4C–4E). Based on the above-mentioned unique structural superiority, the GC-TiO@CHF/S cathode in this work shows a significant improvement on high rate performance over previously reported nanoparticle-type cathode materials.

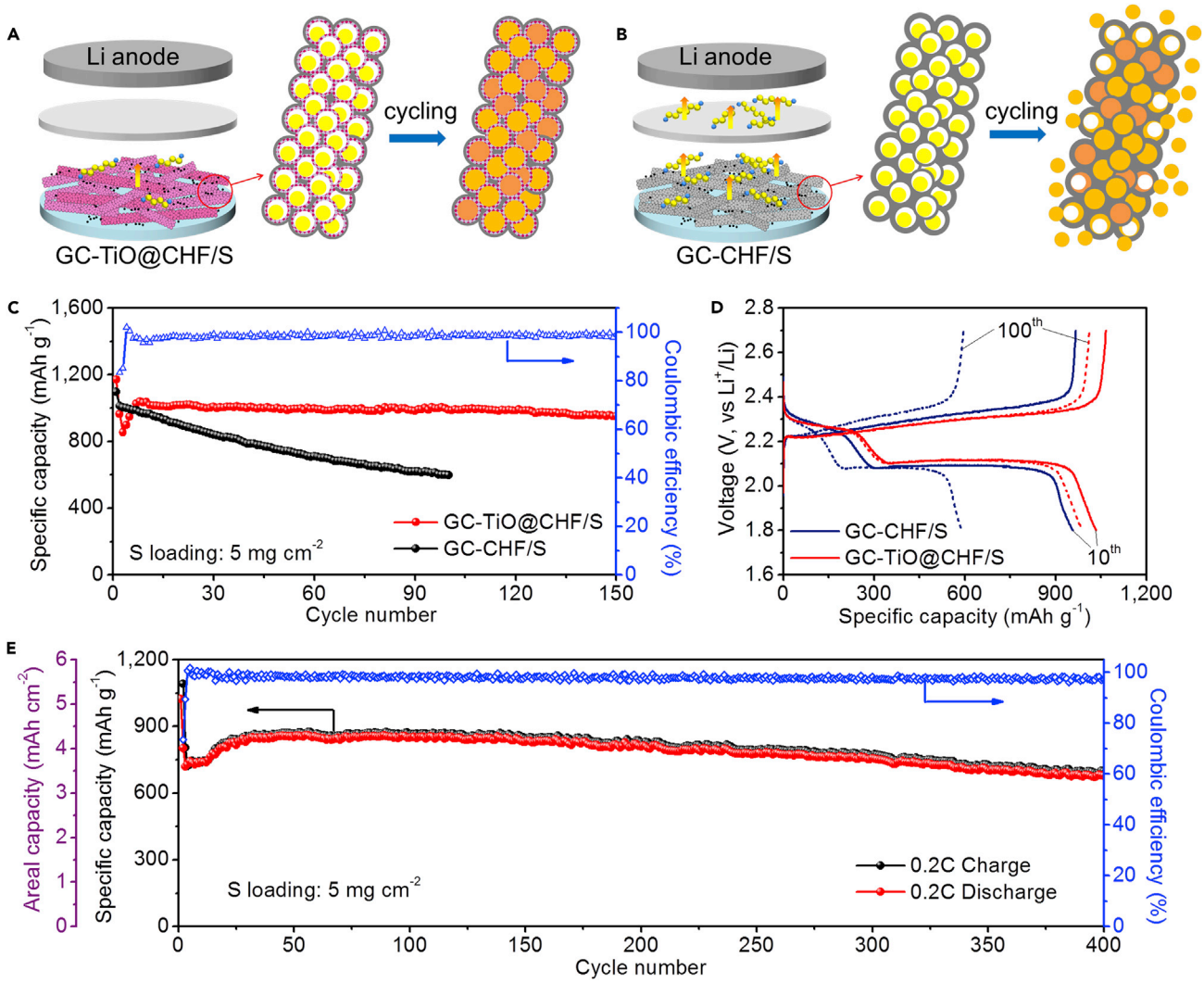


Figure 5. Cycle Performance of the GC-TiO@CHF/S Cathode

(A and B) Schematic illustration of the structural advantage of the GC-TiO@CHF/S cathode (A) compared with GC-CHF/S (B).

(C and D) Cycle properties at 0.1 C (C) and typical voltage profiles (D) of GC-TiO@CHF/S and GC-CHF/S.

(E) Prolonged cycle performance of GC-TiO@CHF/S at 0.2 C.

To demonstrate the significance of TiO in the sulfur host, the cycling performances of GC-TiO@CHF/S and GC-CHF/S were compared in coin cells with the same assembly conditions. The morphologies and structures of the two samples were similar, and the main difference was the presence of TiO in the GC-TiO@CHF/S cathode. In every single redox reaction unit of the GC-TiO@CHF host, there is a layer of TiO anchoring inside the carbon sheath. The metallic conductivity and polar surface of TiO can accelerate the redox kinetics of sulfur species inside the nano-chambers, and synergistically suppress the dissolution of polysulfides by strong chemical adsorption (Figures 5A and S13). The good cycling stability of NP-TiO@C/S further confirms the important role of TiO in the composite structure (Figure S14). In contrast, the pure carbon host can only slow down the dissolution process of polysulfides by relatively weaker physical confinement. Once a high content of polysulfides is generated in the electrode, the solvated polysulfides may easily detach from the electrode surface and diffuse away, leading to

continuous decrease in capacity during the cycling test (Figure 5B). Due to the grape-cluster-like structure, both the GC-TiO@CHF/S and GC-CHF/S electrodes with high sulfur loading deliver high initial capacities of 1,172 and 1,085 mAh g^{-1}, respectively. In the following cycles, GC-TiO@CHF/S exhibits a very stable cycling performance over 150 cycles, and achieves a high retention capacity of 988 mAh g^{-1} at the 100th cycle, which is 95% of the 10th discharge capacity (Figures 5C and 5D). The average Coulombic efficiency of over 150 cycles was calculated to be 98%, which is comparable with similar high-sulfur-loading electrodes;[21,33,35] note that no special interlayer or ceramic-modified separator was applied in our cells. In comparison, the GC-CHF/S cell suffers from rapid capacity decay, retaining a capacity of 591 mAh g^{-1} at the 100th cycle (Figures 5C and 5D), indicating a much more serious degree of polysulfide dissolution into the electrolyte. By checking the morphologies and colors of the cycled cathode electrodes, separators, and lithium metal anodes disassembled from different coin cells, it can be further concluded that the TiO layer can effectively restrict the diffusion of polysulfides from the electrode (Figures S15–S17). The GC-TiO@CHF/S cathode shows good prolonged cycling stability at 0.2 C (Figure 5E). After activation during the first several cycles, the GC-TiO@CHF/S cathode delivers stable capacity over 400 cycles, and specific capacities of 810, 757 and 680 mAh g^{-1} are obtained at the 200th, 300th, and 400th cycles, corresponding to capacity retentions of 79%, 74%, and 66%, respectively. Throughout the cycle life, the areal capacity of the cathode is > 3 mAh cm^{-2}, which is close to the value of state-of-art commercial cathode electrodes of Li-ion batteries. Considering there is still room for large improvement with this prototype GC-TiO@CHF/S electrode, such as further increasing the coating mass and moderately lowering the contents of binder and carbon additives, we are highly optimistic that a Li-S battery with much higher energy density than commercial Li-ion batteries can be achieved.

DISCUSSION

The grape-cluster-like structural design is fundamentally different from previously reported nanoparticle-type cathodes as well as the integral electrodes in lithium-sulfur batteries. The hierarchical structure of GC-TiO@CHF/S successfully inherits several advantages of both nanosized and micrograde materials. First, the electrochemical redox reactions are confined in each nanochamber, enabling excellent charging/discharging kinetics for active sulfur species. Second, the large-sized secondary fibers can be easily slurry coated on an Al foil current collector, forming stable and sufficiently high mass-loading electrode film without using any special binder or conductive additive, which is entirely compatible with conventional battery manufacturing protocols. Third, benefiting from the highly conductive composition of TiO and carbon, the space-efficient packing arrangement, together with the one-dimensional shape with large length-diameter ratio, an efficient conducting network can be easily constructed in the high-mass-loading electrode film, thus promising much higher capacities than simple nanoparticles at various current densities. Fourth, the presence of TiO guarantees excellent cycling stability. Thus, the GC-TiO@CHF/S composite cathode shows high specific/areal capacity, good rate capability, and stable cycle life. Nonetheless, many future developments are still needed to meet the requirements for commercial lithium-sulfur batteries, such as reducing the cost of starting materials for synthesis, further increasing the ratio of active material in the electrode, lowering the usage of electrolyte and protecting the lithium anode. The present work will inspire materials scientists for rational designs of various electrode materials for next-generation batteries.

EXPERIMENTAL PROCEDURES

Synthesis

All chemicals were purchased from Sigma-Aldrich and used without further purification. SiO_2 nanospheres were prepared by a modified Stöber method. Typically, 30 mL of deionized water, 9 mL of ethanol, and 6 mL of ammonia were mixed under stirring, then a mixture of 45 mL of ethanol and 2.55 mL of tetraethyl orthosilicate was added to this solution. After stirring for 4 hr, SiO_2 nanospheres were washed by centrifugation (8,000 rpm, 3 min) with ethanol several times, and the collected SiO_2 nanospheres were re-dispersed in ethanol for subsequent use. The TiO_2 layer was coated on SiO_2 by a method developed by our group.[52] Typically, around 300 mg of SiO_2 was ultrasonically dispersed in a solution containing 49 mL of ethanol, 0.4 g of hexadecylamine, and 1 mL of ammonia. After stirring for 10 min, 0.6 mL of titanium isopropoxide was added dropwise into the dispersion under vigorous stirring. After reaction for 1 hr, SiO_2@TiO_2 was collected and washed by centrifugation with ethanol and dimethylformamide (DMF) several times. Then, as-prepared SiO_2@TiO_2 was ultrasonically dispersed in 2.5 mL of DMF and mixed with 0.2 g of polyacrylonitrile (PAN) under stirring at 60°C overnight. Then, the SiO_2@TiO_2@PAN fibers were electrospun on aluminum foil from the precursor solution. The solution was continuously fed at a rate of 1 mL hr^{-1}; the distance between the collector and the stainless steel needle was fixed at 12 cm, and a voltage of 17 kV was applied. The SiO_2@TiO_2@PAN fibers were first stabilized at 260°C for 2 hr in air, and then annealed at 700°C for 3 hr in N_2. SiO_2 was etched by immersing the sample in 4 wt % HF solution for 8 hr, and then GC-TiO_2@CHF was collected by vacuum filtration. GC-TiO@CHF was prepared by annealing GC-TiO_2@CHF at 1,000°C for 4 hr in an N_2:H_2 (95:5) atmosphere. To synthesize the GC-CHF host, SiO_2 spheres were mixed directly with PAN in DMF solution, and the SiO_2@PAN fibers were electrospun with the same parameters described above. After annealing at 700°C for 3 hr in N_2, and etched in 4 wt % HF solution, the GC-CHF host was obtained. To prepare the sulfur-based composite cathode, GC-TiO@CHF and commercial sulfur powder (1:4, weight ratio) were mixed and sealed in a glass vessel and then heated at 300°C for 4 hr under an N_2 atmosphere in a quartz tube furnace. After cooling down, the GC-TiO@CHF/S composite was obtained. The GC-CHF/S composite was prepared by the same method. Another control group of NP-TiO@C/S was prepared by a previously reported method.[42]

Materials Characterization

Field-emission SEM (JEOL-6700) and TEM (JEOL, JEM-2010) were applied to characterize the morphologies and structures of the samples. EDX spectroscopy attached to TEM (JEOL, JEM-2100F) was used to record the elemental distribution. A Bruker D2 Phaser X-ray diffractometer (Cu Kα radiation, λ = 1.5406 Å) was applied to determine the crystal phases of the samples. TGA (Shimadzu DRG-60) was applied to test the sulfur or carbon contents of the composite samples. GC-TiO@CHF/S was carefully pressed within a cylindrical die at 5 MPa, and then the compaction density was calculated based on the mass and volume of the pressed pellet. Nitrogen sorption isotherms were measured at liquid nitrogen temperature with a Quantachrome Instruments Autosorb AS-6B. The BET-specific surface area was calculated from the adsorption data in the relative pressure range of 0.06–0.2.

Electrochemical Measurements

Electrode films of GC-TiO@CHF/S and GC-CHF/S were prepared by the same method. Typically, 80 wt % of active material, 10 wt % of conductive carbon (Super-P), and 10 wt % of PVDF were well mixed in N-methylpyrrolidone (NMP) to form a stable viscous slurry, and then the slurry was casted onto the aluminum foil

and dried at 70°C overnight. The NP-TiO@C/S electrode film was prepared by the same method with active material, Super-P, and PVDF in the weight ratio of 70:20:10. The areal mass loading of sulfur was about 5.0 mg cm^{-2}. The 2032-type coin cells were assembled using a Celgard 2300 membrane as separator and Li metal as anode. The electrolyte was 1.0 M lithium bis(trifluoromethanesulfonyl) imide (LiTFSI) in 1,3-dioxolane and dimethoxymethane (1:1, volume) with 0.2 M of $LiNO_3$. The cells were cycled in a voltage range of 1.8–2.7 V using a NEWARE battery tester. When the current densities were higher than 0.5 C, the voltage window was slightly expanded to 1.7–2.7 V. The specific capacities and current densities were based on the mass of sulfur. The electrochemical impedance spectroscopy measurements were tested over the frequency range of 100 kHz to 0.1 Hz at open-circuit voltage before initial discharging.

SUPPLEMENTAL INFORMATION

Supplemental Information includes 17 figures and can be found with this article online at http://dx.doi.org/10.1016/j.joule.2017.06.003.

AUTHOR CONTRIBUTIONS

Z.L. and X.W.L. conceived the idea and co-wrote the manuscript. Z.L. carried out the synthesis of materials and the electrochemical evaluation. J.T.Z. and B.Y.G. helped with characterization of the materials. All the authors discussed the results and commented on the manuscript.

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